

MH1301 Discrete Mathematics

Handout 2: Pigeonhole Principle, Permutations and Combinations

We continue with Part I on counting. We learn a seemingly trivial counting principle called the Pigeonhole Principle, which turns out to be a fundamental and useful tool in solving counting problems. We also study in greater detail counting number of arrangements of objects. Finally, we introduce combinatorial proofs, which is a method of proving counting formulas.

- The Pigeonhole Principle
- Permutations
- Combinations
- Combinatorial Proof

Pigeons and Pigeonholes

Motivational Question

Suppose that a flock of 20 pigeons flies into a set of 19 pigeonholes to roost. Prove that at least one of these 19 pigeonholes must have at least two pigeons in it.

Proof: We prove the claim by contradiction.

Suppose not. Then each pigeonhole has at most one pigeon in it.

Then in all 19 pigeonholes, there are at most 19 pigeons.

This is less than the total number of pigeons in the 19 pigeonholes, which is a contradiction. □

Pigeonhole Principle

Theorem 1 (Section 6.2)

The Pigeonhole Principle. Let k be a positive integer. If $k + 1$ or more objects are placed into k boxes, then there is at least one box containing two or more the objects.

$p \Rightarrow q \rightarrow \text{conditional}$

Proof: We prove the Pigeonhole Principle by contraposition, that is, we prove that if none of the k boxes contain more than one object, then less than $k+1$ objects are placed into the k boxes

Suppose that none of the k boxes contains more than one object, then the total number of object would be at most k . Thus less than $k + 1$ objects are placed into the k boxes.



Application

~~Re look idon't understand~~

Example 4 (Section 6.2)

Show that for every integer n , there is a multiple of n that has only 0s and 1s in its decimal expansion.

Proof: Consider the set of $n + 1$ integers

$$a \equiv b(n)$$

$$c \equiv d(n)$$

$$a+d \equiv c+d(n)$$

$$X = \{1, 11, 111, \dots, \underbrace{111\dots1}_{n+1 \text{ digit}}\}.$$

$$5 \equiv 1(4)$$

$$49 \equiv 1(4)$$

$$49 - 5 \equiv 1 - 1(4)$$

Note that there are n possible remainders when an integer is divided by n . By the pigeonhole principle, there must be two elements $x \neq x'$ in X with the same remainder when divided by n .

Assume without loss of generality that $x > x'$, then $x - x'$ is a multiple of n , which has a decimal expansion consisting entirely of 0s and 1s. $\square$

Recall some important functions

Definition

Given any real number x , the **floor of** x , denoted $\lfloor x \rfloor$, is defined as follow

$$\lfloor x \rfloor = \text{that unique integer } n \text{ such that } n \leq x < n+1$$

Symbolically, if x is a real number and n is an integer, then

$$\lfloor x \rfloor = n \iff n \leq x < n+1.$$

$$\lfloor -4.37 \rfloor = -5$$

$$\lceil -4.37 \rceil = -4$$

Definition

Given any real number x , the **ceiling of** x , denoted $\lceil x \rceil$, is defined as follow

$$\lceil x \rceil = \text{that unique integer } n \text{ such that } n-1 < x \leq n$$

Symbolically, if x is a real number and n is an integer, then

$$\lceil x \rceil = n \iff n-1 < x \leq n.$$

Generalized Pigeonhole Principle

Theorem 2 (Section 6.2)

If N objects are placed into k boxes, then there is at least one box containing at least $\lceil N/k \rceil$ objects.

20 pigeons into 7 holes

$$\frac{20}{7} = 2 \dots$$

Proof: Suppose not. Then none of the boxes contains more than $\lceil N/k \rceil - 1$ objects. Then the total number of objects is at most

$$k \left(\left\lceil \frac{N}{k} \right\rceil - 1 \right) < k \left(\left(\frac{N}{k} + 1 \right) - 1 \right) = N,$$

which contradicts with the existence of N objects. □

*from definition of ceiling function,
 $\lceil x \rceil = n \iff n-1 < x \leq n$*

Hair Coincidence

Example

Singapore has more than 5 million non-bald people. Show that there exists 26 people with the same number of hairs on their heads!

(Assume that every person has less than 200,000 hairs.)

Proof:

The number of pigeons here is $N \geq 5,000,001$, and the number of pigeonholes is $k \leq 199,999$, hence

$$\left\lceil \frac{5,000,001}{199,999} \right\rceil = 26.$$

20 students each bought one burger from a choice of chicken, salmon, or vegetarian burger. Label the following with T if the statement is always true and F otherwise.

Waiting for next slide.

At least one student bought a chicken burger. ✓

At most 7 students bought the same type of burger. ✓

At least 6 students bought the same type of burger. ✓

At least 7 students bought the same type of burger. ✓



Permutation

Permutation, r -Permutation

A **permutation** of a set S of distinct elements is an **ordered** arrangement of these elements.

An r -**permutation** of a set S with n elements is an **ordered** selection of r distinct elements from S , where $r \leq n$.

The number of r -permutations of a set of n elements is denoted by $P(n, r)$.

Example

Let $S = \{a, b, c\}$. Give an example of a permutation of S . Find $P(3, 2)$.

Solution:

- (b, c, a) is a permutation of S ,
 - $(a, b), (b, a), (a, c), (c, a), (b, c), (c, b)$ are all 2-permutations of S .
- Hence $P(3, 2) = 6$.



Value of $P(n, r)$

Theorem 1 (Section 6.3)

For any positive integers n and r with $1 \leq r \leq n$, there are

$$P(n, r) = n(n-1)(n-2) \cdots (n-r+1)$$

r -permutations of a set with n distinct elements.

Proof:

- The first entry of the r -permutation has n possibilities.
- The second entry of the r -permutation has $n-1$ possibilities.
- Continuing like this, the r^{th} entry of the r -permutation has $n-(r-1)$ possibilities.
- By the product rule, the number of r -permutations is $n(n-1) \cdots (n-r+1)$.



Value of $P(n, r)$

For $r = 0$, we note that $P(n, 0) = 1$ since there is exactly one way to order zero elements.

Next, recall that $n! = 1 \cdot 2 \cdot 3 \cdots n$.

From Theorem 1, we have immediately the following corollary.

Corollary 1 (Section 6.3)

If n and r are integers such that $0 \leq r \leq n$, then

$$P(n, r) = \frac{n!}{(n-r)!}.$$

Special Cases

- $P(n, n) = n!$.
 - ▶ note that $P(n, n)$ is just the number of permutations of n elements.
 - ▶ remember: $1! = 1$, $2! = 2$, $3! = 6$, $4! = 24$, $5! = 120$, $6! = 720$,
 - ▶ $0!$ is defined to be 1.
- $P(n, 1) = n$.
- $P(n, 0) = 1$.

Example - COMPUTER

- (i) How many ways can the letters in the word COMPUTER be arranged in a row?
- (ii) How many ways can the letters in the word COMPUTER be arranged if the letters CO must be next to each other (in order) as a unit?

Solution:

- (i) COMPUTER contains 8 different characters. So it is equivalent to counting the number of permutation with 8 elements, which is

$$P(8, 8) = 8! = 40320.$$

- (ii) By combining CO together, there are 7 different 'characters': CO, M, P, U, T, E, R. So it is equivalent to counting the number of permutation with 7 elements, which is

$$P(7, 7) = 7! = 5040.$$

Example - Prize Winners

How many ways are there to select a first-prize winner, a second-prize winner and a third-prize winner from 200 different students in a class?

** order matters*

Solution:

It is equivalent to counting number of 3-permutation from a set of 200 elements. Hence

$$\begin{aligned} P(200, 3) &= 200 \cdot 199 \cdot 198 \\ &= 7,880,400. \end{aligned}$$

or

$$\frac{200!}{(200-3)!}$$

Combination

r -Combination

An r -combination of a set S with n elements is an **unordered** selection of r elements from S , where $r \leq n$.

Thus, an r -combination is simply a subset of S with r elements.

The number of r -combinations of a set of n elements is denoted by $C(n, r)$ or $\binom{n}{r}$, is called a **binomial coefficient** and read as “ n choose r ”.

write this instead of nC_r
 nC_r
 nC_r

Example

Let $S = \{a, b, c\}$. Give an example of a 3-combination of S . Find $C(3, 2)$.

Solution:

- $\{a, b, c\}$ is a 3-combination of S .
- $\{a, b\}, \{a, c\}, \{b, c\}$ are all 2-combinations of S . Hence $C(3, 2) = 3$.
Note that $\{a, b\}$ and $\{b, a\}$ are the same combination.

Differences between Permutation and Combination

- Permutations are **ordered**: Both the identities of the selected elements and the order of the selection matter.

For example, (a, b) and (b, a) are two different permutations.

- Combinations are **unordered**: Only the identities of the selected elements matter.

For example, $\{a, b\}$ and $\{b, a\}$ are the same combination.

Relation between Permutation and Combination

Motivational Question.

What is the value of $C(n, r)$?

To help us answer this question, let us first explore the case when $n = 3$.
Let $S = \{a, b, c\}$.

- $\{a, b\}, \{a, c\}, \{b, c\}$ are all 2-combinations of S . Hence $C(3, 2) = 3$.
- $(a, b), (b, a), (a, c), (c, a), (b, c), (c, b)$ are all 2-permutations of S .
Hence $P(3, 2) = 6$.

Relation between Permutation and Combination

Motivational Question

Is there a connection between $P(3, 2)$ and $C(3, 2)$?

The construction of a 2-permutation of S can be thought of as a two-step process.

- (1) Choose a set of 2 elements from S , in which there are $C(3, 2)$ ways.
- (2) Choose an order for these selected subset, which has $2!$ ways since there are 2 elements.

By the product rule, $P(3, 2) = C(3, 2) \cdot 2!$

$$\iff C(3, 2) = \frac{P(3, 2)}{2!}.$$

Is there a similar connection between $P(n, r)$ and $C(n, r)$?

Relation between Permutation and Combination

How to construct an r -permutation?

- (1) Choose a set of r elements from S , in which there are $C(n, r)$ ways.
- (2) Choose an order for these selected subset, which has $r!$ ways since there are r elements.

Consequently, $P(n, r) = C(n, r) \cdot P(r, r)$. It implies that

$$C(n, r) = \frac{P(n, r)}{P(r, r)} = \frac{n!/(n-r)!}{\cancel{r!}} = \frac{n!}{r!(n-r)!}.$$

We record this result in the following theorem on $C(n, r)$.

$C(n, r)$ and special Cases

Theorem 2 (Section 6.3)

For any integers r and n with $0 \leq r \leq n$,

$$C(n, r) = \frac{n!}{r!(n-r)!}.$$

Special Cases

- $C(n, n) = 1$. (Recall that $0! = 1$.)
- $C(n, 1) = n$.
- $C(n, n-1) = n$.
- $C(n, 0) = 1$.

Example - Strings

Example 14 (Section 6.3)

How many 0-1 bit strings of length n contain exactly r 1's?

Solution: The positions of r 1's in a bit string of length n form an r -combination of the set $\{1, 2, \dots, n\}$.

Hence, there are $C(n, r)$ bit strings of length n that contain exactly r 1's.

↪ out of n bits choose r of them



Example - Game

Example 15 (Section 6.3)

There are 9 boys and 11 girls. How many ways are there to choose 3 boys and 4 girls to form a team?

Solution:

- There are $C(9, 3) = \frac{9!}{3!6!} = 84$ ways to choose 3 boys from 9 boys.
- There are $C(11, 4) = \frac{11!}{4!7!} = 330$ ways to choose 4 girls from 11 girls.
- By the product rule, the final answer is

$$C(9, 3) \cdot C(11, 4) = 84 \cdot 330 = 27,720.$$

Example - Team

Suppose two members A and B of a group of 12 people insist on working as a pair – any team must contain either both or neither. How many 5-person teams can be formed?

10 others & A and B

Solution: There are two possibilities:

- Both A and B are in the team: there are $C(10, 3) = \frac{10!}{3!7!} = 120$ ways.
- Neither A nor B is in the team: there are $C(10, 5) = \frac{10!}{5!5!} = 252$ ways.

By the sum rule, the total number of possible teams is

$$C(10, 3) + C(10, 5) = 120 + 252 = 372.$$

no overlaps

Example - Powerball!



In the Powerball lottery there are five regular numbers on white balls, and one special number on a red ball, the “Powerball”. The five regular numbers must be **distinct integers** between 1 and 69. The sixth number is between 1 and 26; it does not have to be distinct from the other five. You win the jackpot if you match all six numbers

Example - Powerball!

How many possible Powerball ticket combinations are there?

Solution:

- There are $C(69, 5)$ choices to choose the white balls.
- There are 26 choices to choose the “Powerball”.
- By the product rule, there are $C(69, 5) \cdot 26 = 292,201,338$ possible combinations.

In the week of 13 Jan, 2016, Jackpot prize hit \$1.5 billion, then won by 3 winners with a split of \$500 million each.



Example - Cards

- (i) How many poker hands of five cards can be dealt from a standard deck of 52 cards?
- (ii) Also, how many ways are there to select 47 cards from a standard deck of 52 cards?

Solution:

- (i) Since the order in which the five cards are dealt from a deck of 52 cards does not matter, there are

$$C(52, 5) = \frac{52!}{5!47!} = 2,598,960$$

different hands of five cards that can be dealt.

- (ii) Similarly, there are $C(52, 47) = \frac{52!}{47!5!} = 2,598,960$ ways to select 47 cards from a standard deck of 52 cards.

Corollary

Corollary 2 (Section 6.3)

Let n and r be nonnegative integers with $r \leq n$. Then

$$C(n, r) = C(n, n - r).$$

1st Proof: From the r -combination theorem, we have

$$C(n, r) = \frac{n!}{r!(n-r)!} = \frac{n!}{(n-r)![n-(n-r)]!} = C(n, n-r).$$

□

2nd Proof: Suppose S is a set with n elements. Every subset A of S with r elements corresponds to a subset of S with $n - r$ elements, namely $\bar{A}$ (the complement of A with respect to S).

Consequently, $C(n, r) = C(n, n - r)$

□

Combinatorial Proof

Definition 1 (Section 6.3)

A **combinatorial proof** of an identity is a proof that uses counting arguments to prove that both sides of the identity count the same objects but in different ways.

Example

Prove that for all $n \geq k \geq 1$,

no. of ways to get k objects out of n $\swarrow$

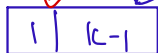
$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

See Handout 3 for properly written proof

Solution:



Case 1:



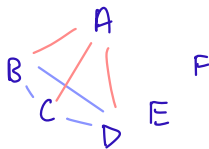
Case 2:



$$\text{no. of ways } S = \binom{1}{1} \binom{n-1}{k-1} + \binom{1}{0} \binom{n-1}{k} = \binom{n}{k}$$

Example

Among a party of six, prove that there are 3 people who either know one another or are strangers to one another.



— know each other
— stranger

5 lines between A & B, C, D, E, F (Pigeons)
and two types of lines (Pigeon holes)

By Gen. Pigeonhole Principle, there are at least $\lceil \frac{5}{2} \rceil = 3$

lines of the same type, say know each other
B and C are strangers. For otherwise A-B-C know one another and we are done. Likewise, B and D are strangers, and so are C & D.
 $\therefore$ B, C, D are strangers to one another.

Example

Among a party of n people, $n \geq 2$, prove that there exist two people knowing the same number of other people.

Case 1: Somebody knows everybody (e.g. A)

$\therefore$ For every body in the party of n people, $\overset{\text{pigeon}}{\text{no. of people anyone knows is in the range}}$
 $\underbrace{1, 2, 3, \dots, n-1}_{n-1 \text{ (pigeonholes)}}$

Case 2: Nobody knows everybody

The range is $\underbrace{0, 1, \dots, n-2}_{n-1 \text{ pigeon holes}}$ $\overset{\text{pigeons}}{\text{for } n \text{ people}}$

Example

Let n be a positive integer. Prove that the number of ways of expressing

$$n = a + b \quad \text{where } a \text{ and } b \text{ are integers with } a \geq b \geq 0$$

is equal to the number of ways of expressing

$$n = c + 2d \quad \text{where } c \text{ and } d \text{ are nonnegative integers.}$$

$$(c, d) \rightarrow (c+d, d)$$

$$\begin{aligned} 7 &= 7+0 \\ &= 6+1 \\ &= 5+2 \\ &= 4+3 \end{aligned}$$

$$(a, b)$$

$$\begin{aligned} &= 7+0 \\ &= 5+2 \cdot 1 \\ &= 3+2 \times 2 \\ &= 1+3 \times 2 \end{aligned}$$

$$(c, d)$$

$$\begin{array}{cc} \text{then } c+d \geq d \geq 0 \\ \uparrow \quad \uparrow \\ a \quad b \end{array}$$

~~Go~~ Go watch again
→ before toilet break

Example

Let

$$\text{Pf: Let } T = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid n = a + b, \quad a \geq b \geq 0\} \quad \text{and}$$
$$S = \{(c, d) \in \mathbb{Z} \times \mathbb{Z} \mid n = c + 2d, \quad c, d \geq 0\}.$$

is e

consider $\varphi: S \rightarrow T$

$$\varphi(c, d) = (c + d, d).$$

Since $c + d + d = n$ and $c + d \geq d \geq 0$, we have $(c + d, d) \in T$.

Also, clearly φ is well-defined.

φ is 1-1: Suppose $\varphi(c_1, d_1) = \varphi(c_2, d_2)$. Then $c_1 + d_1 = c_2 + d_2$
and $d_1 = d_2$. $\therefore$ we also have $c_1 = c_2$. $\therefore (c_1, d_1) = (c_2, d_2)$.

φ is onto: Let $(a, b) \in T$. Since $a \geq b \geq 0$, we have $a - b \geq 0$ and
so $(a - b, b) \in S$. Clearly $\varphi(a - b, b) = (a, b)$.

$\therefore \varphi$ is a bijection and so $|S| = |T|$.

g

Reading Assignment

Rosen (7th edition):

- Chapter 6.2, The Pigeonhole Principle
- Chapter 6.3, Permutations and Combinations